

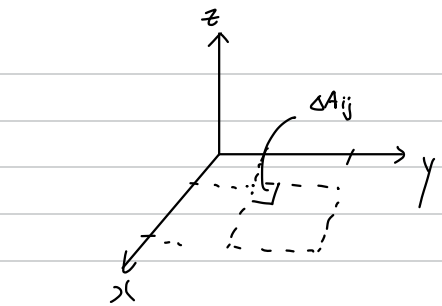
Double Integrals (15.1 - 15.4)

For Rectangular Regions:
Precise definition:

$$1. \iint_R f(x,y) dA = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij} \quad \text{for equal partitions}$$

$$2. \iint_R f(x,y) dA = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij} \quad \|P\| = \max(\Delta x_{ij})$$

"biggest partition $\rightarrow 0$ "



Physical significance:

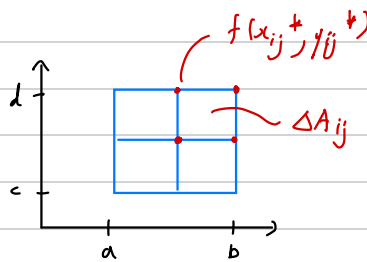
Volume under surface $z = f(x,y)$ in region $x \in [a,b], y \in [c,d]$

Estimation:

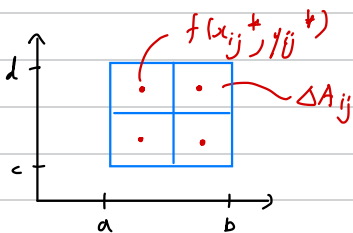
Using double Riemann sum:

$$V \approx \sum_{i=1}^n \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

Typically Pick $n=n$, ΔA_{ij} to be equal,
we can pick any corner of the rectangles



Using Midpoint Rule:



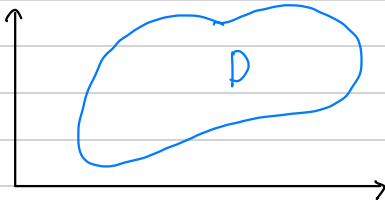
Fubini's Theorem

$$\iint_R f(x,y) dA = \int_a^b \int_c^d f(x,y) dy dx = \int_c^d \int_a^b f(x,y) dx dy \quad \text{for constants } a, b, c, d$$

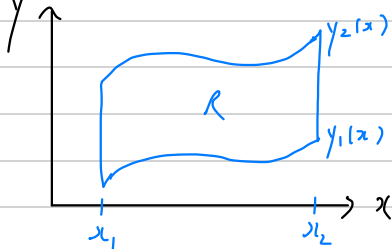
thus if $f(x,y) = g(x)h(y)$

$$\int_c^d \int_a^b g(x)h(y) dx dy = \left(\int_a^b g(x) dx \right) \left(\int_c^d h(y) dy \right)$$

Over general Regions:

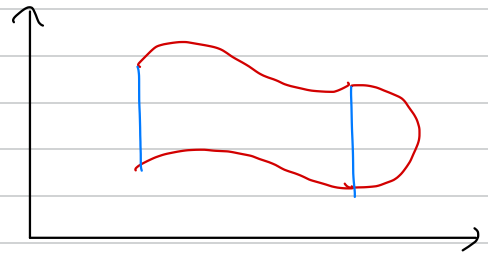


Type 1 (between two functions of $y, y(x)$)

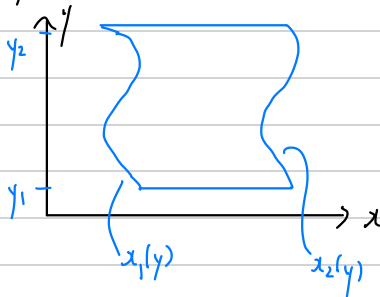


$$\int_{x_1}^{x_2} \int_{y_1(x)}^{y_2(x)} f(x,y) dy dx$$

Mixed Region



Type 2 (between two functions of $x, x(y)$)

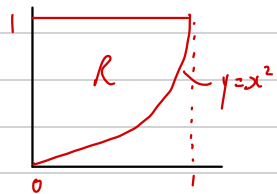


$$\int_{y_1}^{y_2} \int_{x_1(y)}^{x_2(y)} f(x,y) dx dy$$

Divide into type 1 and 2

Changing the order integration

Example:



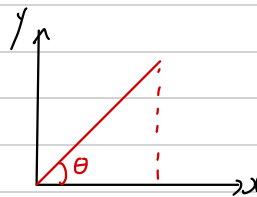
$$\begin{aligned} \int_0^1 \int_{x^2}^1 f dy dx \\ = \int_0^1 \int_0^{\sqrt{y}} f dx dy \end{aligned}$$

Property:

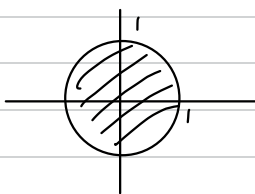
$$\iint_R 1 dA = \text{Area of Region } R$$

Double integral in Polar coords

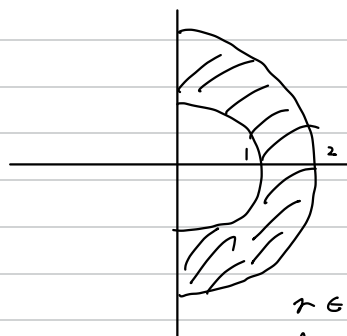
$$\begin{aligned} x &= r \cos \theta, \quad y = r \sin \theta \\ r^2 &= x^2 + y^2 \\ \theta &= \tan^{-1} \frac{y}{x} \end{aligned}$$



Fixed regions



$$\begin{aligned} r &\in [0, 1] \\ \theta &\in [0, 2\pi] \end{aligned}$$



$$\begin{aligned} r &\in [1, 2] \\ \theta &\in [-\pi/2, \pi/2] \end{aligned}$$

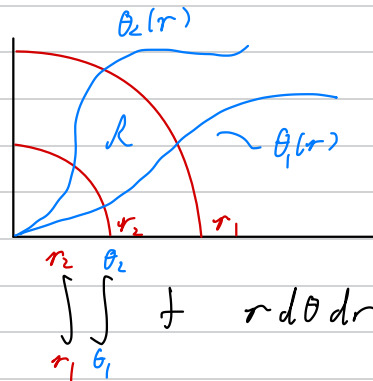
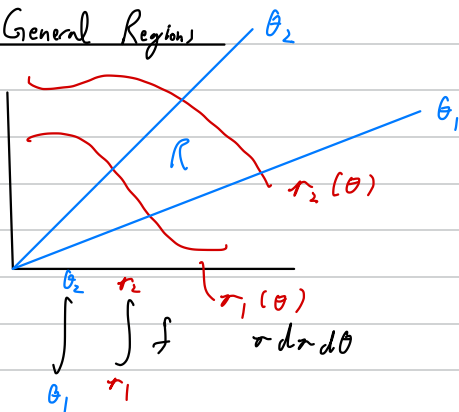
Precise definition:

$$\lim_{n, m \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m g(r_i^*, \theta_i^*) \Delta r \Delta \theta, \text{ where } g(r, \theta) = r f(r \cos \theta, r \sin \theta)$$

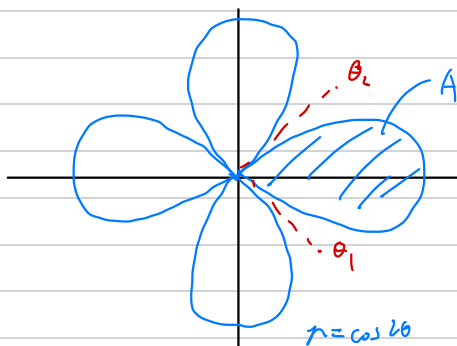
Conversion

$$\iint_R f(x, y) dA = \iint_{R'} f(r \cos \theta, r \sin \theta) r dr d\theta$$

General Region



Polar graphs and double integrals



Step 1: Find intersection \$\theta\$
Step 2: \$\int_{\theta_1}^{\theta_2} \int_0^{\cos 2\theta} r dr d\theta\$

Roscs:

$$r = a \cos n\theta$$

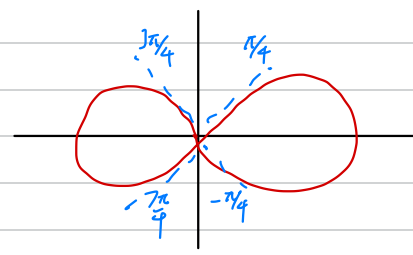
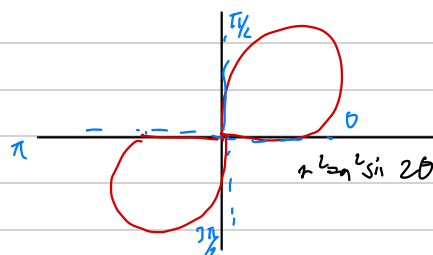
\$n\$ even

\$\hookrightarrow 2n\$ leaved

\$n\$ odd

\$\hookrightarrow n\$ leaved

Lemniscates



$$r^2 = a^2 \cos 2\theta$$

Application of Double Integrals

Density and Mass

If $\rho = \rho(x, y)$

$$m = \iint_D \rho \, dA$$

Moment

M_x / Moment about x -axis:

$$M_x = \iint_D y \rho(x, y) \, dA$$

$$M_y = \iint_D x \rho(x, y) \, dA$$

Center of mass

$$m\bar{x} = M_y, \quad m\bar{y} = M_x$$

$$\bar{y} = \iint_D y \rho(x, y) \, dA \cdot \left(\iint_D \rho \, dA \right)^{-1}$$

$$\bar{x} = \iint_D x \rho(x, y) \, dA \cdot \left(\iint_D \rho \, dA \right)^{-1}$$

Moment of Inertia

$$I = mr^2$$

$$I_x = \iint_D y^2 \rho(x, y) \, dA \quad \rightarrow \quad I_o = I_x + I_y = \iint_D (x^2 + y^2) \rho(x, y) \, dA$$

$$I_y = \iint_D x^2 \rho(x, y) \, dA$$

Surface Area (15.5)

$$A = \iint_D \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1} \, dA$$

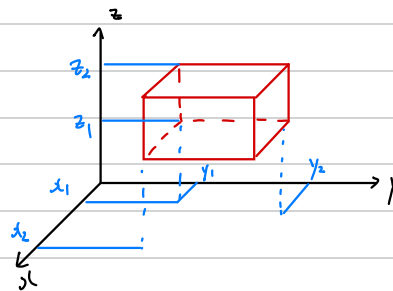
Triple Integrals (15.6 - 15.9)

For a rectangular box:

Precke Definition:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^p f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

$$\int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} f \, dx \, dy \, dz$$



For a general region

$$\iiint f(x, y, z) \, dz \, dA$$

$$\int_{x_1}^{x_2} \int_{y_1(x)}^{y_2(x)} \int_{z_1(x, y)}^{z_2(x, y)} f \, dz \, dy \, dx$$

Application of triple integrals:

Volume:

$$\iiint_E dV$$

Moment

$$M_{yz} = \iiint_E x \rho \, dV$$

CM

$$\bar{x} = \frac{\iiint_E x \rho \, dV}{m} = \frac{M_{yz}}{m}$$

Moment of inertia

$$I_x = \iiint_E (y^2 + z^2) \rho \, dV$$

Mass:

$$\iiint_E \rho \, dV$$

$$M_{xz} = \iiint_E y \rho \, dV$$

$$M_{yz} = \iiint_E x \rho \, dV$$

$$\bar{y} = \frac{\iiint_E y \rho \, dV}{m} = \frac{M_{xz}}{m}$$

$$\bar{z} = \frac{\iiint_E z \rho \, dV}{m} = \frac{M_{xy}}{m}$$

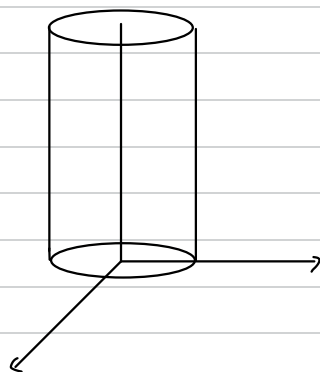
$$I_y = \iiint_E (x^2 + z^2) \rho \, dV$$

$$I_z = \iiint_E (y^2 + x^2) \rho \, dV$$

↳ Centroid is when ρ is constant

$$I_0 = \iiint_E (x^2 + y^2 + z^2) \rho \, dV$$

Cylindrical Coordinates

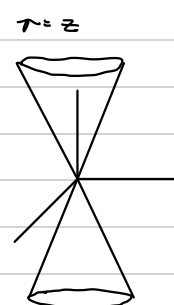
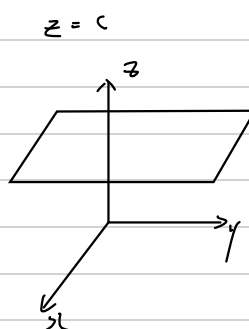
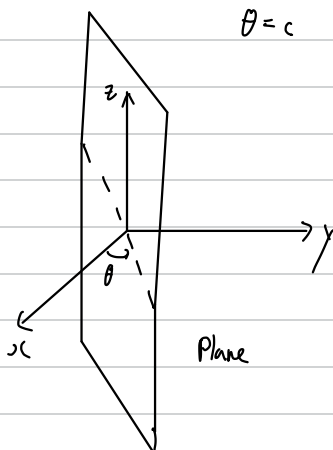
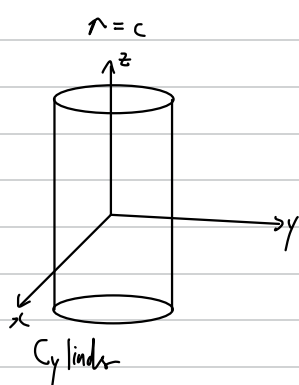


$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

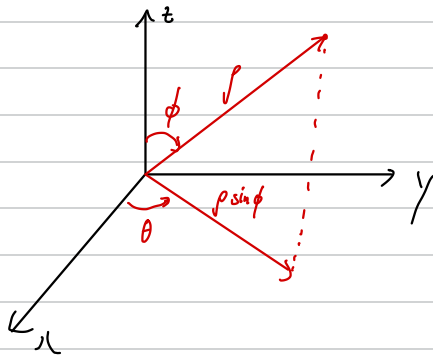
Example objects:



Conversion

$$\iiint_E f(x, y, z) \, dV \rightarrow \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta) \, r \, dz \, dr \, d\theta$$

Spherical Coordinates



$$\begin{aligned}x &= \rho \sin \phi \cos \theta \\y &= \rho \sin \phi \sin \theta \\z &= \rho \cos \phi\end{aligned}$$

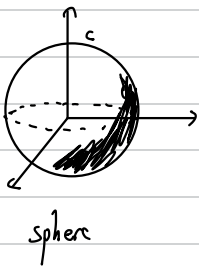
$$\rho^2 = x^2 + y^2 + z^2$$

$$0 < \theta < 2\pi$$

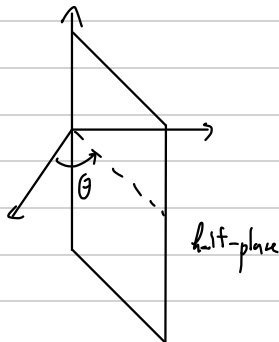
$$0 < \phi < \pi$$

Example objects

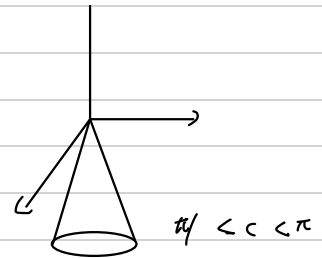
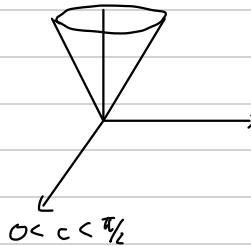
$$\rho = c$$



$$\theta = c$$



$$\phi = c$$



Conversion

$$\iiint_E f(x, y, z) dV \longrightarrow \int_{\phi_1}^{\phi_2} \int_{\theta(\phi)}^{\theta(\phi)} \int_{\rho(\theta, \phi)}^{\rho(\theta, \phi)} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi$$

Change of variables

2D:

$$\begin{aligned}\text{Let } x &= g(u, v) \\ y &= h(u, v)\end{aligned}$$

The Jacobian, $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = x_u y_v - x_v y_u$

$$\text{and } \int_{u_1}^{u_2} \int_{v_1}^{v_2} f(x, y) \underbrace{dy dx}_{dA} = \int_{u_1}^{u_2} \int_{v_1}^{v_2} f(g(u, v), h(u, v)) \underbrace{\left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv}_{dA}$$

For 3D:

$$\begin{aligned}\text{Let } x &= g(u, v, w) \\ y &= h(u, v, w) \\ z &= k(u, v, w)\end{aligned}$$

$$\int_{u_1}^{u_2} \int_{v_1}^{v_2} \int_{w_1}^{w_2} f(x, y, z) dx dy dz = \int_{u_1}^{u_2} \int_{v_1}^{v_2} \int_{w_1}^{w_2} f(g(u, v, w), h(u, v, w), k(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

Taylor Expansion in multiple variables (Notes 1.9)

2 variable

First form:

1st order

2nd order

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx \left(f(x_0, y_0) + \left(f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y \right) + \frac{1}{2!} \left(f_{xx}(x_0, y_0) \Delta x^2 + 2f_{xy}(x_0, y_0) \Delta x \Delta y + f_{yy}(x_0, y_0) \Delta y^2 \right) \dots \right)$$

(let $x_0 + \Delta x$ be x , $y_0 + \Delta y$ be y)

Second form:

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0) \cdot (x - x_0) + f_y(x_0, y_0) \cdot (y - y_0) + \frac{1}{2!} \left(f_{xx}(x_0, y_0) (x - x_0)^2 + 2f_{xy}(x_0, y_0) (x - x_0)(y - y_0) + f_{yy}(x_0, y_0) (y - y_0)^2 \right)$$

3 variable

Linear approximation:

$$f(x, y, z) \approx f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

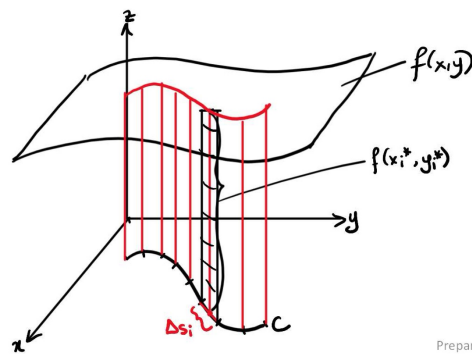
Line Integrals (16.2 - 16.3)

2D:

Definition:

$$\int_C f(x, y) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$

$$= \int_{t_1}^{t_2} f(x(t), y(t)) \underbrace{\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}}_{ds} dt$$



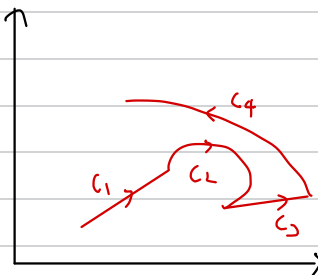
$$\int_C ds = \text{arc length.}$$

3D:

$$\int_C f(x, y, z) ds = \int_{t_1}^{t_2} f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Piecewise smooth curve

$$\int_C f(x, y) ds = \int_{C_1} f ds + \int_{C_2} f ds + \dots$$



$$\int_C f(x, y) ds = \int_C f(x, y) ds$$

Mass, CM of wires

$$M = \int_C \rho(x,y) ds$$

$$M_x = \int_C \rho y ds \longrightarrow \bar{x} = \frac{M_y}{M} = \frac{\int_C \rho x ds}{M}$$

$$M_y = \int_C \rho x ds \longrightarrow \bar{y} = \frac{M_x}{M} = \frac{\int_C \rho y ds}{M}$$

Line integrals wrt x, y

$$\int_C f(x,y) dx = \int_{t_1}^{t_2} f(x(t), y(t)) x'(t) dt$$

$$\int_C f(x,y) dy = \int_{t_1}^{t_2} f(x(t), y(t)) y'(t) dt$$

Parametrizing curves

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1 \quad \text{for a straight line}$$

$$\vec{r}_0 = (x_0, y_0, z_0)$$

$$\vec{r}_1 = (x_1, y_1, z_1)$$

$$x = r \cos t, \quad y = r \sin t \quad \text{for a circle}$$

Line integrals over vector fields

$$W = \int_C \vec{F} \cdot \vec{T} ds = \int_C \vec{F} \cdot d\vec{r} \quad \begin{array}{l} \vec{F} \rightarrow \text{force function} \\ \vec{T} \rightarrow \text{unit tangent function} \\ \vec{r}(t) \rightarrow \text{parametrized curve } C \end{array}$$
$$d\vec{r} = \vec{T}(ds)$$
$$= \int_C \frac{\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) ds}{|\vec{r}'(t)|}$$

$$= \int_{t_1}^{t_2} \vec{F}(\vec{r}(t)) \cdot \frac{\vec{r}'(t)}{|\vec{r}'(t)|} |\vec{r}'(t)| dt$$

$$= \int_{t_1}^{t_2} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \quad \text{Evaluation method 1}$$

$$= \int P dx + Q dy + R dz, \quad \vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$$

Note

$$\int_C \vec{F} \cdot d\vec{r} = - \int_{-C} \vec{F} \cdot d\vec{r}$$

Fundamental Theorem of the integral

If F is a conservative vector field,
 $F = \nabla f$, f is the potential function

then

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)), \text{ for } b \leq t \leq a$$

Results

If any of the below is true, all other statements must also be true:

- ↳ $\oint_C \vec{F} \cdot d\vec{r} = 0$
- ↳ $\int_C \vec{F} \cdot d\vec{r}$ is independent of path taken
- ↳ $\vec{F}$ is conservative

Potential function:

2D:

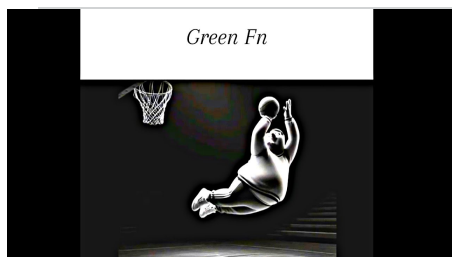
If $\vec{F} = P\hat{i} + Q\hat{j}$, by Clairaut's Theorem,
 if $\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$, then f exists

3D:

If $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ and

$$\left. \begin{aligned} f_{xy} &= \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \\ f_{xz} &= \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x} \\ f_{yz} &= \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y} \end{aligned} \right\} \text{ then } f \text{ exists}$$

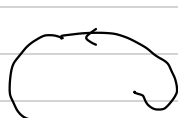
Green's theorem (16.4)



"Closedness"

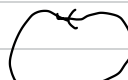


closed!



not closed

"Simple"



simple



not simple

"Orientation"



+ve



-ve

Does not intersect itself except at its endpoints

Counterclockwise → +ve
 Clockwise → -ve

Theorem:

Let C be positively oriented, piecewise-smooth, simple, closed curve in the plane and let D be the region bounded by C . If P and Q have cont partial derivatives on an open region that contains D , then

$$\oint_C P dx + Q dy$$

$$= \int_C \vec{F} \cdot d\vec{r}, \quad \vec{F} = P\hat{i} + Q\hat{j}$$

Other forms of green's theorem:

$$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{\partial D} P dx + Q dy = \iint_D (\text{curl } \vec{F}) \cdot \vec{n} dA$$

unit vector

$$\oint_C \vec{F} \cdot \vec{n} ds = \iint_D \text{div } \vec{F}(x,y) dA$$

Parametric Surface and their Surface Areas (16.6)

For the surface $\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$, surfaces have 2 parameters, curves have 1

We can parametrize by letting:

$$x = x(u,v) \quad y = y(u,v) \quad z = z(u,v)$$

typically,

$$x = u$$

$$y = v$$

$$z = f(u,v)$$

Tangent Plane

For a surface @ position $\vec{r}(u_0, v_0)$

The Tangent Plane contains the vectors:

$$\vec{r}_u(u_0, v_0) = \frac{\partial x}{\partial u} \hat{i} + \frac{\partial y}{\partial u} \hat{j} + \frac{\partial z}{\partial u} \hat{k}$$

$$\vec{r}_v(u_0, v_0) = \frac{\partial x}{\partial v} \hat{i} + \frac{\partial y}{\partial v} \hat{j} + \frac{\partial z}{\partial v} \hat{k}$$

and the normal vector, $\vec{n} = \vec{r}_u \times \vec{r}_v$

Equation of plane: $(x-x_0, y-y_0, z-z_0) \cdot \vec{n} = 0$

Surface Area

$$A(S) = \iint_D |\vec{r}_u \times \vec{r}_v| dA, \quad (u,v) \in D$$

Using Green's Theorem, $\iint_D dA = A_{\text{area}}$

then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$, we can pick any $\frac{\partial Q}{\partial x}$ and $\frac{\partial P}{\partial y}$

If $x=x, y=y, z=f(x,y)$

$$A(S) = \iint_D \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1} dA$$

then

$$A = \iint_D dA = \oint_C P dx + Q dy \quad \text{for} \quad \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

Surface Integrals (16.7)

$$\iint_S f(x, y, z) dS$$

$$= \iint_D f(\vec{r}(u, v)) |\vec{r}_u \times \vec{r}_v| du dv, \quad u, v \in D$$

$$\text{If } x=u, \quad y=v, \quad z=g(x, y)$$

$$\text{then } dS = \left(\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 + 1 \right) dA \quad \text{and}$$

$$f(\vec{r}(x, y)) = f(x, y, g(x, y))$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 + 1} dA$$

Smooth surfaces
 $\vec{r}_u \times \vec{r}_v \neq \vec{0}$

Oriented Surfaces

2-sided surface

(So for only a Möbius strip is mentioned to be non-orientable)

unit normal vectors can point in 2 directions
 $\vec{n}_1 = -\vec{n}_2$

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \quad \text{or} \quad \frac{\vec{r}_v \times \vec{r}_u}{|\vec{r}_u \times \vec{r}_v|}$$

Surface integral over vector field, Flux

$$\iint_S \vec{F} \cdot d\vec{S}$$

$$= \iint_D \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) du dv, \quad u, v \in D$$

it opposite orientation, multiply by -1

Curl and Divergence (16.5)

$$\text{If } \vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$$

Curl

$$\text{curl } \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}$$

$$\text{Since } \nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

$$\text{curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{array}{c} \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \\ \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{array} \begin{array}{c} \times \\ \times \\ \times \\ \times \end{array} \begin{array}{c} Q \\ R \\ P \\ Q \end{array} \longrightarrow \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \dots$$

Divergence

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$
$$= \nabla \cdot \vec{F}$$

Properties of Curl, Divergence

$\nabla \times (\nabla f) = 0$, if $\operatorname{curl} \vec{F} = 0 \rightarrow \mathbf{F}$ is a conservative vector field

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \rightarrow \operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\operatorname{div} (\nabla f) = \nabla^2 f$$

Stokes' Theorem (16.8)

Boundary curve, the orientation

$$\left(\int_C \vec{F} \cdot d\vec{r} \right) = \iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \iint_S \operatorname{curl} \vec{F} \cdot \vec{n} \, dS = \iint_D \operatorname{curl} \vec{F} \cdot \vec{n} \, |r_u \times r_v| \, du \, dv$$

line integral around a boundary curve of S

surface integral of S

$|r_u \times r_v| \, du \, dv$

Divergence Theorem (16.9)

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F} \, dV$$

boundary surface of E

solid region